HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2012)

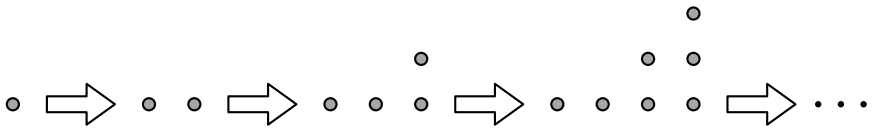
Student No.: 10 Name: _____ Marks: / 52

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, the 1st pattern consists of 1 dot. For any positive integer n, the $(n+1)$th pattern is formed by adding n dots to the nth pattern. Find the number of dots in the 8th pattern.



A. 22

B. 29

C. 36

D. 37

(DSE-2012 PII Q12)

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2. The rectangular coordinates of the point P are $(-3, -3\sqrt{3})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(3, 150°)$.

B. $(3, 330°)$.

C. $(6, 150°)$.

D. $(6, 330°)$.

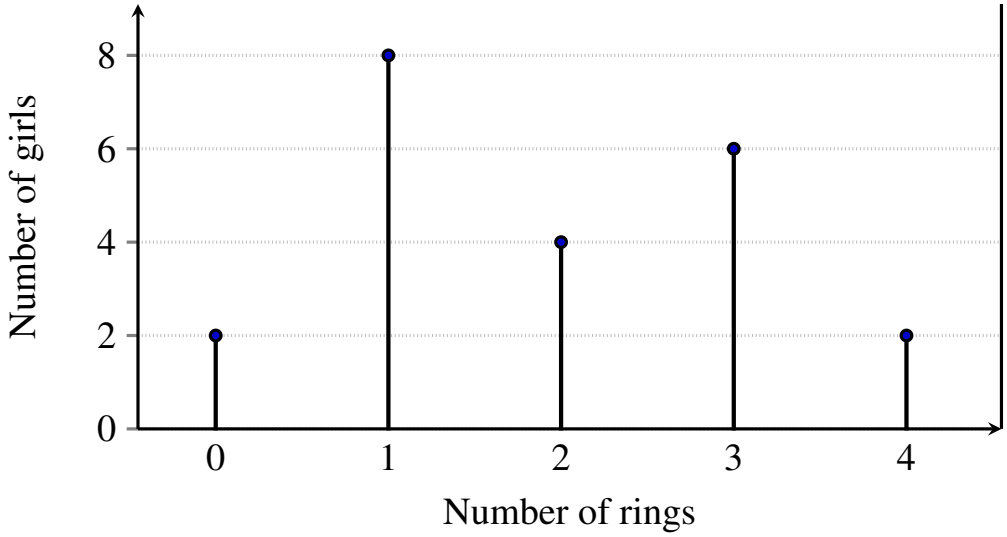
(DSE-2012 PII Q23)

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3. The bar chart below shows the distribution of the numbers of rings owned by the girls in a group. Find the standard deviation of the distribution correct to 2 decimal places.



A. 1.04

B. 1.16

C. 1.19

D. 2.09

(DSE-2012 PII Q29)

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4. $AD000002012_{16} =$

A. $(10)16^{11} + (13)16^{10} + 8210.$

B. $(10)16^{12} + (13)16^{11} + 131360.$

C. $(11)16^{11} + (14)16^{10} + 8210.$

D. $(11)16^{12} + (14)16^{11} + 131360.$

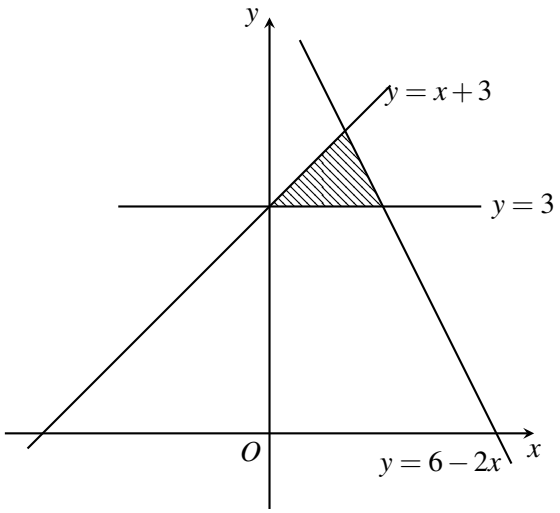
(DSE-2012 PII Q33)

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5. The figure shows a shaded region (including the boundary). If (h,k) is a point lying in the shaded region, which of the following are true?



I. $k \geq 3$

II. $h - k \geq -3$

III. $2h + k \leq 6$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2012 PII Q36)

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6. A drama club is formed by 12 boys and 8 girls. If a team of 5 students is selected from the club to participate in a competition and the team consists of at least one girl, how many different teams can be formed?

A. 3960

B. 14712

C. 15448

D. 15504

(DSE-2012 PII Q43)

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7. A box contains six balls numbered 7, 8, 8, 9, 9 and 9 respectively. John repeats drawing one ball at a time randomly from the box without replacement until the number drawn is 9. Find the probability that he needs exactly three draws.

A. $\frac{1}{2}$

B. $\frac{1}{6}$

C. $\frac{1}{8}$

D. $\frac{3}{20}$

(DSE-2012 PII Q44)

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8. The daily wage of Ada is 20% higher than that of Billy while the daily wage of Billy is 20% lower than that of Christine. It is given that the daily wage of Billy is $480.

(a) Find the daily wage of Ada.

(b) Who has the highest daily wage? Explain your answer.

(4 marks) (DSE-2012 PI Q04)

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9. (a) Find the range of values of x which satisfy both $\frac{4x+6}{7} > 2(x-3)$ and $2x-10 \leq 0$.

(b) How many positive integers satisfy both the inequalities in (a)?

(4 marks) (DSE-2012 PI Q06)

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10. Let \$C be the cost of painting a can of surface area A m^2. It is given that C is the sum of two parts, one part is a constant and the other part varies as A. When $A = 2$, $C = 62$; when $A = 6$, $C = 74$.

(a) Find the cost of painting a can of surface area 13 m^2.

(b) There is a larger can which is similar to the can described in (a). If the volume of the larger can is 8 times that of the can described in (a), find the cost of painting the larger can.

(6 marks) (DSE-2012 PI Q11)

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11. The standard deviation of the test scores obtained by a class of students in a Mathematics test is 10 marks. All the students fail in the test, so the test score of each student is adjusted such that each score is increased by 20% and then extra 5 marks are added.

(a) Find the standard deviation of the test scores after the score adjustment.

(b) Is there any change in the standard score of each student due to the score adjustment? Explain your answer.

(3 marks) (DSE-2012 PI Q15)

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12. Figure 5(a) shows a right pyramid $VABCD$ with a square base, where $\angle VAB = 72°$. The length of a side of the base is 20 cm. Let P and Q be the points lying on VA and VD respectively such that PQ is parallel to BC and $\angle PBA = 60°$. A geometric model is made by cutting off the pyramid $VPBCQ$ from $VABCD$ as shown in Figure 5(b).

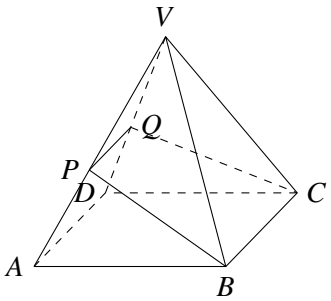


Figure 5(a)

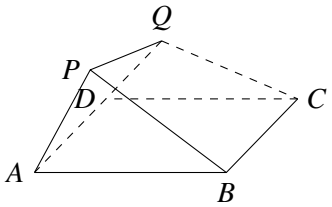


Figure 5(b)

(a) Find the length of AP.

(b) Let α be the angle between the plane $PBCQ$ and the base $ABCD$.

(i) Find α.

(ii) Let β be the angle between PB and the base $ABCD$. Which one of α and β is greater? Explain your answer.

(8 marks) (DSE-2012 PI Q18)

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13. In a city, the air cargo terminal X of an airport handles goods of weight $A(n)$ tonnes in the nth year since the start of its operation, where n is a positive integer. It is given that $A(n) = ab^{2n}$, where a and b are positive constants. It is found that the weights of the goods handled by X in the 1st year and the 2nd year since the start of its operation are 254 100 tonnes and 307 461 tonnes respectively.

(a) (i) Find a and b. Hence find the weight of the goods handled by X in the 4th year since the start of its operation.

(ii) Express, in terms of n, the total weight of the goods handled by X in the first n years since the start of its operation.

(b) The air cargo terminal Y starts to operate since X has been operated for 4 years. Let $B(m)$ tonnes be the weight of the goods handled by Y in the mth year since the start of its operation, where m is a positive integer. It is given that $B(m) = 2ab^m$.

(i) The manager of the airport claims that after Y has been operated, the weight of the goods handled by Y is less than that handled by X in each year. Do you agree? Explain your answer.

(ii) The supervisor of the airport thinks that when the total weight of the goods handled by X and Y since the start of the operation of X exceeds 20 000 000 tonnes, new facilities should be installed to maintain the efficiency of the air cargo terminals. According to the supervisor, in which year since the start of the operation of X should the new facilities be installed?

(13 marks) (DSE-2012 PI Q19)

I can finish _____ questions

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